

Chapter 04: Laplace Transforms & Operational Calculus

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Course: Applied Ordinary Differential Equations (ENGR 213) | Textbook: Dennis G. Zill (7th Ed., Chapter 4)

1. Executive Summary & First-Principles Philosophy

The **Laplace Transform** is an operational integral transform that converts differential equations in the time domain (t) into algebraic equations in the frequency / complex s -domain.

Instead of dealing with calculus operations (derivatives and integrals), we perform high school algebra, solve for $Y(s)$, and then apply the **Inverse Laplace Transform** via partial fractions to return to $y(t)$. Initial conditions are embedded directly into the transform step!

2. Core Mechanics & Mathematical Engine

A. Formal Definition

Given a function $f(t)$ defined for $t \geq 0$:

$$\mathcal{L}f(t) = F(s) = \int_0^\infty e^{-st} f(t) dt$$

B. Master Transform Table of Elementary Functions

Time Domain $f(t)$	Laplace Domain $F(s)$	Convergence Region
1	$\frac{1}{s}$	$s > 0$
$t^n \ (n \in \mathbb{Z}^+)$	$\frac{n!}{s^{n+1}}$	$s > 0$
e^{at}	$\frac{1}{s-a}$	$s > a$
$\sin(kt)$	$\frac{k}{s^2+k^2}$	$s > 0$
$\cos(kt)$	$\frac{s}{s^2+k^2}$	$s > 0$
$\sinh(kt)$	$\frac{k}{s^2-k^2}$	$s > k$
$\cosh(kt)$	$\frac{s}{s^2-k^2}$	$s > k$

C. Operational Theorems

1. First Translation (Frequency Shift) Theorem:

$$\mathcal{L}e^{at}f(t) = F(s - a)$$

2. Transforms of Derivatives:

$$\mathcal{L}f'(t) = sF(s) - f(0)$$

$$\mathcal{L}f''(t) = s^2F(s) - sf(0) - f'(0)$$

$$\mathcal{L}f^{(n)}(t) = s^nF(s) - s^{n-1}f(0) - \dots - f^{(n-1)}(0)$$

3. Fully Worked Exam Archetype: Solving a 2nd-Order IVP

Problem: Solve using Laplace transforms:

$$y'' - 4y' + 4y = e^{2t}, \quad y(0) = 0, \quad y'(0) = 1$$

Step-by-Step Solution:

- **Step 1: Take Laplace transform of both sides:**

$$\mathcal{L}y'' - 4\mathcal{L}y' + 4\mathcal{L}y = \mathcal{L}e^{2t}$$

$$(s^2Y(s) - sy(0) - y'(0)) - 4(sY(s) - y(0)) + 4Y(s) = \frac{1}{s - 2}$$

- **Step 2: Substitute initial values $y(0) = 0, y'(0) = 1$:**

$$s^2Y(s) - 1 - 4sY(s) + 4Y(s) = \frac{1}{s - 2}$$

$$(s^2 - 4s + 4)Y(s) - 1 = \frac{1}{s - 2}$$

$$(s - 2)^2Y(s) = 1 + \frac{1}{s - 2} = \frac{s - 1}{s - 2}$$

- **Step 3: Solve algebraically for $Y(s)$:**

$$Y(s) = \frac{s - 1}{(s - 2)^3}$$

- **Step 4: Decompose via Partial Fractions / Shift Theorem:** Rewrite numerator in terms of $(s - 2)$:

$$s - 1 = (s - 2) + 1$$

$$Y(s) = \frac{(s - 2) + 1}{(s - 2)^3} = \frac{1}{(s - 2)^2} + \frac{1}{(s - 2)^3}$$

- **Step 5: Apply Inverse Transform:** By the First Translation Theorem, $\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2}\right\} = t e^{2t}$. For the second term: $\mathcal{L}\{t^2\} = \frac{2!}{s^3} \implies \mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} = \frac{1}{2}t^2$. Therefore, $\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^3}\right\} = \frac{1}{2}t^2 e^{2t}$.

$$y(t) = t e^{2t} + \frac{1}{2} t^2 e^{2t} = e^{2t} \left(t + \frac{1}{2} t^2 \right)$$

4. Rapid Exam Traps & Red Flags

-  **Trap 1: Dropping Negative Signs in Derivative Formulas:** The formula is $s^2 Y - s y(0) - y'(0)$. Both initial condition terms carry **negative signs**.
-  **Trap 2: Completing the Square in Denominators:** For terms like $\frac{s}{s^2+4s+13} = \frac{s}{(s+2)^2+9}$, split the numerator into $(s+2) - 2$ to apply cos and sin shift rules correctly.